

Lesson 4: Direct Hit

GameMaker

VIDEO GAME CREATION

This week: detecting when things touch. Your job is to add one enemy to the screen and make your bullet destroy it on contact - the first piece of your capstone's actual combat loop.

Open your own [YourName]Game project from Lesson 3 - don't start a new one. Everything this week builds on the player and bullet you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder enemy image (enemy_invader_red.png) and a new sound effect (sfx_hit.wav) - you don't need to find or make your own art today.

Setup

1. Open your [YourName]Game project from Lesson 3.
2. Create a new sprite spr_enemy by importing enemy_invader_red.png from this lesson's Assets folder.
3. Create a new object obj_enemy, assign spr_enemy to it.
4. Drag one obj_enemy instance into your room near the top of the play area.

Level 1 - Minimal (MVP)

A single enemy sits on screen. When your bullet touches it, both the bullet and the enemy disappear. This is a complete finish for today - reaching this is enough to move on to next week.

5. On obj_bullet, add a Collision event with obj_enemy (Add Event > Collision > obj_enemy) with the code below - it destroys both the enemy it hit and itself.
6. Press F5 (Run) and shoot the enemy to test - both the bullet and the enemy should disappear on contact.

```
// obj_bullet's Collision event with obj_enemy:
instance_destroy(other);
instance_destroy(self);
```

Alternate Path: GML Visual (no typing)

The same collision-detection pattern works with drag-and-drop blocks instead of typed code. Pick whichever path feels right for you - both are graded the same.

7. On obj_bullet, add a Collision event with obj_enemy, same as the code path.
8. From the Instance category, drag two "Instance - Destroy Instance" blocks into it - set the first block's Instance dropdown to Other, and the second to Self.

9. Press F5 (Run) and test the same way as the code path.

Level 3's ideas map onto blocks the same way, favoring the most specific named block available: "Play Audio" (Audio category) for sound, and "Set Instance Variable" -> Image Blend for the enemy's own visual idea - Image Blend is a recognized built-in on Set Instance Variable's own dropdown, so prefer it over the generic Assign Variable/Set Variable.

Level 2 - Complete (Alpha/Beta)

The enemy is destroyed cleanly and exactly once, even if you fire several bullets in a row - no errors, no lingering references, and a bullet that misses still behaves exactly like it did last week.

- Fire several bullets in a row, including some that miss - each hit destroys exactly one bullet and the enemy, and any bullet that misses keeps flying and destroys itself off-screen exactly like it did in Lesson 3.
- GameMaker's Collision event already only fires once per pair of instances actually overlapping that step, and instance_destroy() is safe to call on an instance that's already been destroyed earlier in the same step (it's simply ignored) - so even the rare case of two bullets hitting the same enemy on the same step doesn't error.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: add sfx_hit.wav as a sound resource (keep its default name, sfx_hit), then call audio_play_sound(sfx_hit, 1, false) at the top of the Collision event, before either instance_destroy() call.
- Visual: give obj_enemy its own image_blend color in its Create event, distinct from the bullet's yellow, so it clearly reads as a threat rather than another shot.
- Structure: obj_bullet's Collision event should only ever detect a hit and clean up (destroy the enemy, destroy itself, play the sound) - anything about how the enemy looks on its own belongs in obj_enemy's own Create event, not bolted onto obj_bullet.

If Something's Wrong

- Nothing happens on contact: confirm the Collision event on obj_bullet is specifically targeting obj_enemy (Add Event > Collision lets you pick which object), not left as a generic collision.
- Errors mentioning other: other only has a meaningful value inside a Collision event - pasting this code into the Step event instead will error or silently do nothing.
- Bullet destroys the enemy but nothing seems to change: confirm obj_enemy was actually placed in the room (an object with zero instances can never collide with anything), and that instance_destroy(other) is really inside the Collision event.

Want More?

Student Resource List > GameMaker section, or gamemaker.io/en/tutorials